STRAIGHTENING LAWS IN ALGEBRAIC COMPLEXITY

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Abstract.

1 Introduction

Many problems in combinatorics, geometry, and computer science can be expressed in terms of the computation of some polynomials. For example, finding a perfect matching in a graph is equivalent to determining if it's *Tutte Matrix* has determinant identically zero. With this, we see that it is important to study the complexity of computing polynomials in general, and specifically computing special polynomials such as the determinant or permanent polynomials. Algebraic complexity theory is the study of the complexity of such computations.

1.1 Lower Bounds and Polynomial Identity Testing

1.2 Geometric Complexity Theory

1.3 Ideals in Algebraic Complexity

2 Preliminaries on Algebraic Complexity

2.1 Computational Models

Definition 2.1 (Algebraic Circuits):

Definition 2.2 (Oracle Circuits):

Definition 2.3 (Algebraic Formulas):

Definition 2.4 (Algebraic Branching Programs):

2.2 Basic Structural Results

Lemma 2.5 (Interpolation):

Lemma 2.6 (Isolation Lemma):

Lemma 2.7 (Isolation Lemma for Monomials):

2.3 Border Complexity

3 Preliminaries on Straightening Laws

3.1 Determinantal Ideals

3.2 Pfaffian Ideals

3.3 Symmetric Determinantal Ideals

3.4 Permanental Ideals

3.5 Hankel Ideals

4 Prior Work

4.1 Generalities on Complexity in Ideals

4.1.1 Principal Ideals

4.1.2 More Than One Generator

4.2 Approximative Results

5 Current Work on Ideals with Straightening Laws

6 Future Work and Open Problems

6.1 Symmetric Determinantal Ideals

6.2 Permanental Ideals

6.3 Hankel and Toeplitz Ideals

6.4 Complexity of Straightening